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Assignment: Version Control with Git and GitHub

Aim: Create a version control account on GitHub and use Git commands to create a repository and push code to GitHub.

Step 1: Create a GitHub Account

1. Go to <https://github.com> in your web browser.
2. Click on "Sign up" and enter your email, create a password, and choose a username.
3. Verify your email address and complete the account setup steps.
4. Once verified, log in to your new GitHub account.

Step 2: Install and Configure Git

```
# Check if Git is already installed
git --version

# Set your username and email (used in every commit)
git config --global user.name "Abhijeet Krushna Topale"
git config --global user.email "your-email@example.com"

# View current configuration
git config --list
```

Step 3: Create a New Repository on GitHub

1. Log in to GitHub and click the "+" icon in the top-right corner, then select "New repository".
2. Enter a repository name (e.g., my-project), add an optional description.
3. Choose Public or Private, and click "Create repository".
4. Copy the repository URL shown on the next page (HTTPS or SSH).

Step 4: Initialize a Local Repository

```
# Create a project folder and move into it
mkdir my-project
cd my-project

# Initialize an empty Git repository
git init

# Check the current status
git status
```

Step 5: Add Files and Make the First Commit

```
# Add all files to the staging area
git add .

# Add a specific file instead (alternative)
git add filename.ext

# Commit the staged changes with a message
git commit -m "Initial commit"

# View commit history
git log
```

Step 6: Connect the Local Repository to GitHub

```
# Link the local repo to the GitHub repository
git remote add origin https://github.com/username/my-project.git

# Verify the remote was added
git remote -v

# Rename default branch to main (if required)
git branch -M main
```

Step 7: Push Code to GitHub

```
# Push local commits to the GitHub repository
git push -u origin main

# Enter GitHub username and Personal Access Token (PAT) when prompted
```

Step 8: Making Further Changes

```
# After editing files, repeat the add-commit-push cycle
git add .
git commit -m "Updated code"
git push origin main

# Pull latest changes from GitHub before working (good practice)
git pull origin main
```

Step 9: Useful Additional Commands

git clone <repository-url>	# Copy a remote repo locally
git branch	# List branches
git branch <branch-name>	# Create a new branch
git checkout <branch-name>	# Switch to a branch
git checkout -b <branch-name>	# Create and switch to a new branch
git merge <branch-name>	# Merge a branch into current branch

```
git diff
git rm <file>
```

```
# Show unstaged changes
# Remove a file and stage the removal
```

Conclusion

A GitHub account was successfully created and a local Git repository was initialized, connected to a remote GitHub repository, and the project code was pushed successfully using Git commands.

GITHUB Account

[LINK](https://github.com/AbhiTopale)

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